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Project 2

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Prepared by :
1. Brendan Chia Yan Fei (A23CS0211)
2. Woo Cheng Shuan (A23CS0283)
3. Cheryl Cheong Kah Voon (A23CS0060)

Section : 02

Lecturer : DR. ROZILAWATI BINTI DOLLAH @ MD. ZAIN

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1.0 Introduction

1.1 Project Overview

The goal of this project is to investigate and use data mining methods, particularly classification algorithms, to extract useful information from the chosen dataset. The dataset includes structured data related to a certain business domain that was previously used in an earlier assignment. For each of the three unique business insights, the project's main goal is to create, train and assess a unique categorization algorithm. Data preparation, model training, assessment and interpretation are all steps in the process. The project shows how several machine learning models may be used to address practical issues by matching each algorithm with a specific business topic. A comparison evaluation of the model's performances and how effectively it aids in corporate decision-making are included in the final product.

1.2 Objectives

- To apply one classification algorithm for each business insight, tailored to the nature of the problem
- To perform necessary data preprocessing, including cleaning, transformation and sampling to ensure data quality
- To develop, train and evaluate three classification models using suitable performance metrics
- To interpret the outcomes of each model and explain how the results provide value or recommendations for business decision-making
- To document the full process in a clear, technical report supported with visualizations and analysis

1.3 Business Questions/Insights Targeted

1.3.1 Prevalence of Diabetes in the Population

Analysis of the BRFSS 2015 dataset shows that around 29.7% of respondents are diagnosed with diabetes or pre-diabetes. This highlights a significant public health concern. Early detection and lifestyle-based prevention strategies are essential to reduce future cases and healthcare costs. This insight supports the development of targeted health interventions and resource allocation for high-risk groups.

1.3.2 Impact of Lifestyle Factors on Health Outcomes

Lifestyle behaviors such as smoking, physical activity, and diet significantly influence health outcomes. Analysis shows that 44.3% of respondents smoke, while 75.6% engage in regular physical activity and 63.4% consume fruits daily—highlighting both healthy habits and areas of concern. These insights underscore the need for targeted public health interventions, including smoking cessation, fitness programs, and nutrition education, to reduce chronic disease risks and improve overall population health.

1.3.3 Predictive Modeling for At-Risk Individuals

Analysis shows that key health features such as high blood pressure, cholesterol levels, BMI, smoking status, and exercise frequency can effectively identify individuals at high risk of diabetes. Although the overall accuracy is slightly lower compared to other methods, this approach demonstrates better sensitivity in detecting potential high-risk cases, making it valuable for early identification. Applying this predictive model in healthcare or corporate wellness programs allows for immediate risk assessment and personalized health recommendations after users complete health questionnaires, thereby improving screening efficiency and optimizing preventive care.

2.0 Dataset Description

2.1 Source of Dataset

The Behavioral Risk Factor Surveillance System (BRFSS) 2015, a nationwide health survey carried out by the Centers for Disease Control and Prevention (CDC) in the United States, served as the dataset for this study. Through telephone interviews, it gathers data on chronic health issues, health related risk behaviors and utilization of preventative services.

2.2 Description of Features

The original dataset, `diabetes_012_health_indicators_BRFSS2015.csv`, includes over 400,000 records and the 22 features in these following 6 categories: health status, lifestyle, healthcare access, medical history, demographics and target.

2.3 Target Variable

There is 2-class categorical variable:

0 – No diabetes

1 – Diabetes

2.4 Summary Statistics

The cleaned dataset (`dmafterprocess.csv`) retains all original features but has undergone the following cleaning steps which is removal of missing or inconsistent data, standardization or transformation of certain variables and class balancing to improve model performance.

3.0 Data Preprocessing

Our group used KNIME to perform data preprocessing

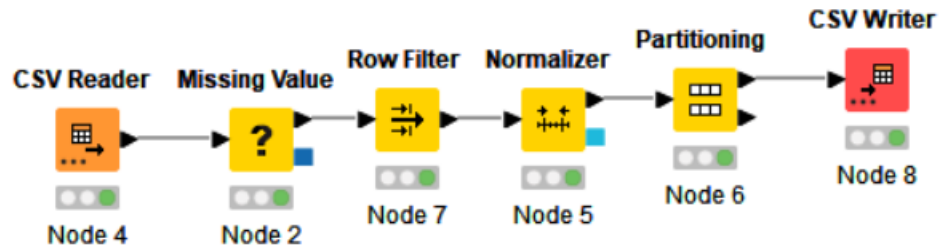


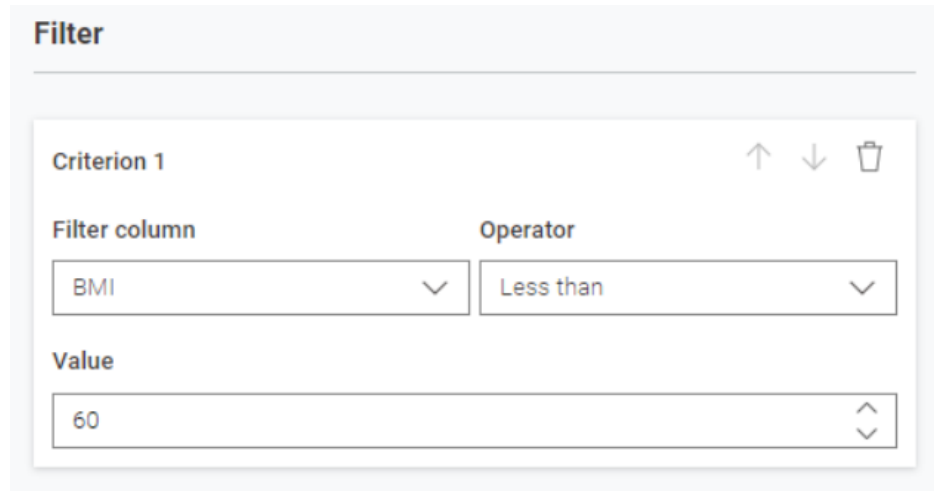
Figure 1: Data Preprocessing using KNIME Analytics Platform

3.1 Missing Value Handling

No missing values were present in the original dataset (diabetes_012_health_indicators_BRFSS2015.csv), so no imputation or removal was necessary. This was confirmed during the initial data inspection phase.

3.2 Row Filtering

Some entries had illogical or extreme values, particularly in the BMI column, with values reaching up to 98, which are unrealistic. To maintain data quality, all rows with BMI > 60 were removed. These entries appeared to be erroneous or random, so filtering helped avoid model distortion.



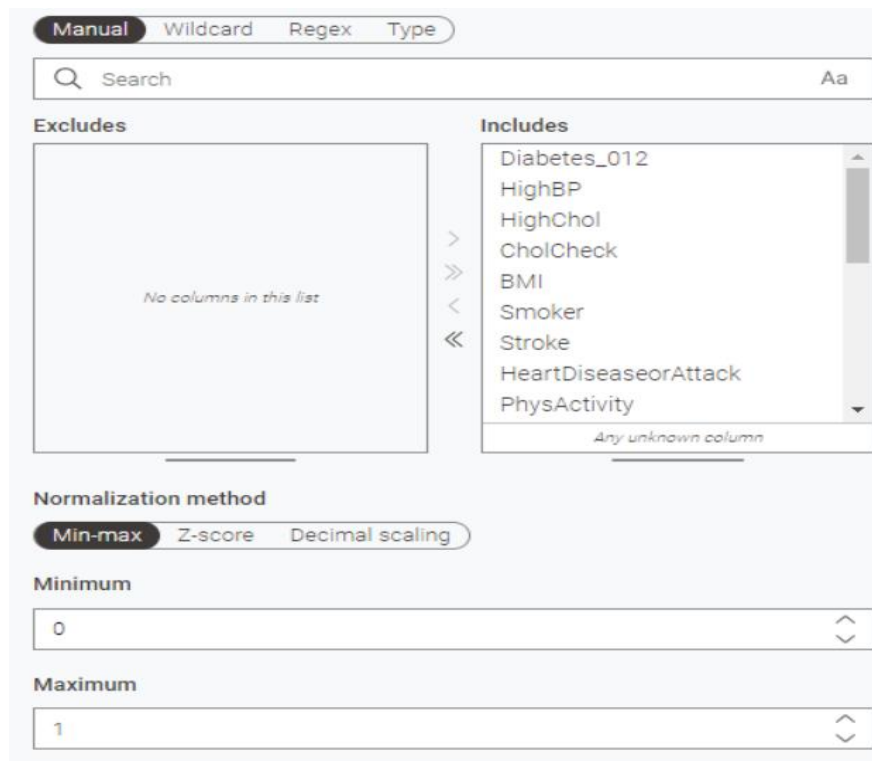
The image shows the 'Filter' widget configuration in KNIME. It has a title 'Filter' and a subtitle 'Criterion 1'. There are three icons at the top right: an up arrow, a down arrow, and a trash can. Below these are two dropdown menus: 'Filter column' with 'BMI' selected and 'Operator' with 'Less than' selected. At the bottom is a 'Value' input field with '60' entered.

Filter column	Operator	Value
BMI	Less than	60

Figure 2: Row filtering in KNIME Analytics Platform

3.3 Normalization

All numerical features (such as BMI, Age, Income, GenHlth, etc.) were normalized to a $[0, 1]$ range using Min-Max scaling. This normalization ensures that features contribute equally to distance-based clustering algorithms like K-Means and DBSCAN, which are sensitive to differences in scale.



The image shows the 'Normalization' widget configuration in KNIME. It has a title 'Manual' and a subtitle 'Wildcard Regex Type'. There is a search bar with 'Aa' on the right. Below the search bar are two lists: 'Excludes' (empty) and 'Includes' (containing Diabetes_012, HighBP, HighChol, CholCheck, BMI, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity). Below the lists are three tabs: 'Min-max' (selected), 'Z-score', and 'Decimal scaling'. At the bottom are two input fields: 'Minimum' with '0' and 'Maximum' with '1'.

Excludes	Includes
No columns in this list	Diabetes_012 HighBP HighChol CholCheck BMI Smoker Stroke HeartDiseaseorAttack PhysActivity

Normalization method: Min-max (selected), Z-score, Decimal scaling

Minimum: 0

Maximum: 1

Figure 3: Normalization in KNIME Analytics Platform

3.4 Data Partitioning

The preprocessed dataset represents an 80% training subset of the full dataset. A random 80/20 split was applied in which 80% of data for training and modeling and 20% reserved for testing. This split was stratified to ensure proportional representation of all classes in the target variable Diabetes_012.

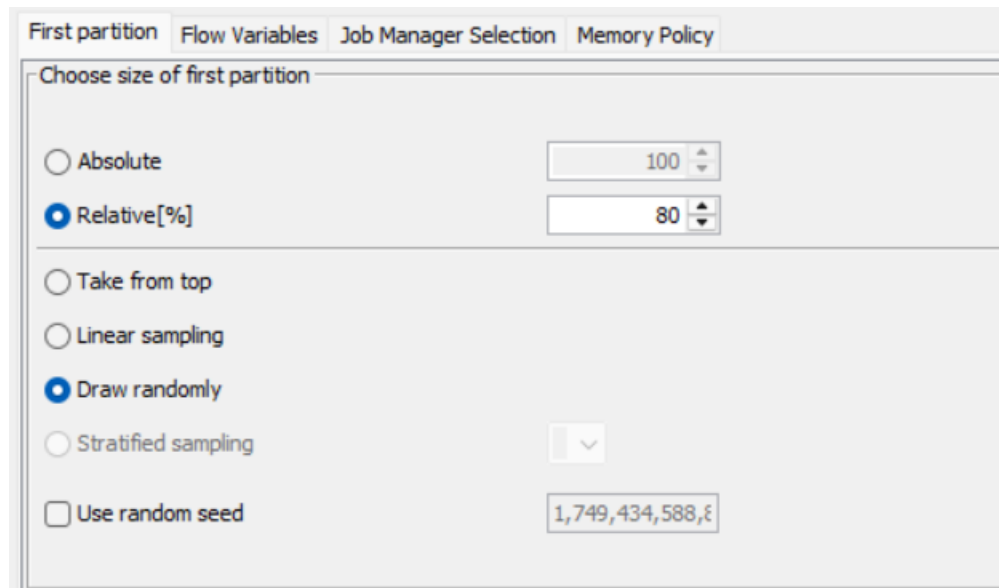


Figure 4: Data Partitioning in KNIME Analytics Platform

3.5 Data Discretization

Although data discretization is typically used to convert continuous features into categorical bins, it was not applied in this project because the dataset primarily consists of binary or already scaled numerical features. Also, all continuous features fall within $[0, 1]$ after normalization.

4.0 Clustering algorithms and Methodology

This section outlines the clustering methods used to analyze the dataset and extract meaningful business insights related to health outcomes and diabetes and diabetes prevalence. In this study, two unsupervised clustering algorithms were applied, which is K-Means Clustering and DBSCAN (Density-Based Spatial Clustering of Application with Noise). To perform clustering using K-Means and DBSCAN, the first 10,000 records from the full dataset of 202,249 entries were selected. This subset was used to reduce computational demands and allow for faster experimentation, especially since DBSCAN has a time complexity of $O(n^2)$, which makes it inefficient for very large datasets.

In this analysis, clustering algorithms **K-Means** and **DBSCAN** are applied to group individuals based on patterns in the dataset. The objective is to analyze how diabetes prevalence varies across the clusters formed by each method. For every cluster, the following metrics are computed:

- **Count**, representing the total number of individuals in the cluster,
- **Total**, indicating the number of diabetic individuals within the cluster, and
- **Prevalence (%)**, calculated as: $Prevalence (\%) = (Count_{Total}) \times 100$

4.1 Insight 1: Impact of Lifestyle Factors on Health Outcomes

To explore the impact of lifestyle factors on health outcomes, feature selection was guided by the relevance of attributes to the business insight. The selected columns, *PhyScActivity*, *Fruits*, *Veggies*, *HvyAlcoholConsump*, *AnyHealthcare*, *GenHlth*, *MentHlth*, *PhysHlth*, and *DiffWalk* reflect behaviors and conditions that directly influence a person's health status. These features were used for clustering to identify patterns and group individuals with similar lifestyle habits and health conditions, enabling more targeted health interventions.

K-Means Clustering

The Elbow Method was used to determine the optimal number of clusters (k) for K-Means clustering. In the plot, the x-axis represents the number of clusters, while the y-axis shows the inertia (within-cluster sum of squares), which measures how internally coherent the clusters are.

As the number of clusters increases, the inertia decreases because data points are grouped into tighter clusters.

The key point in this method is to identify the “elbow”, the point at which the rate of decrease in inertia slows down significantly. In this case, the elbow is observed at $k = 3$.

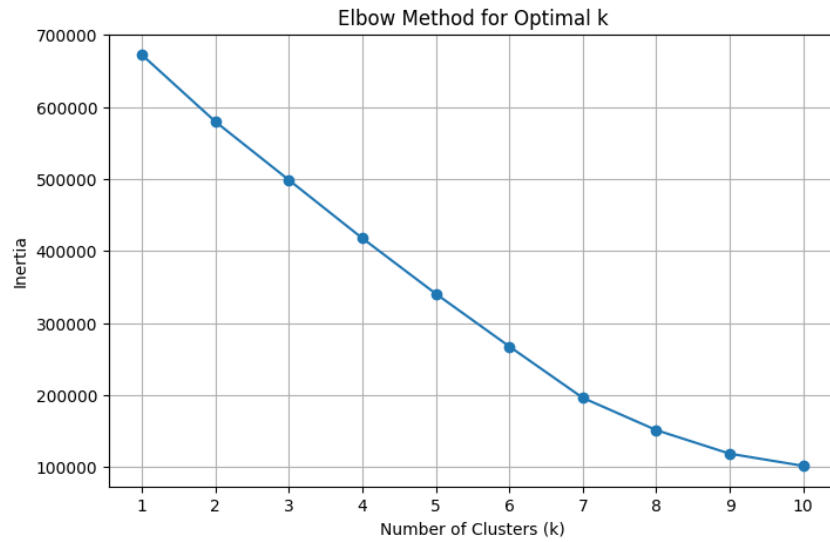


Figure: Elbow Method for Optimal k

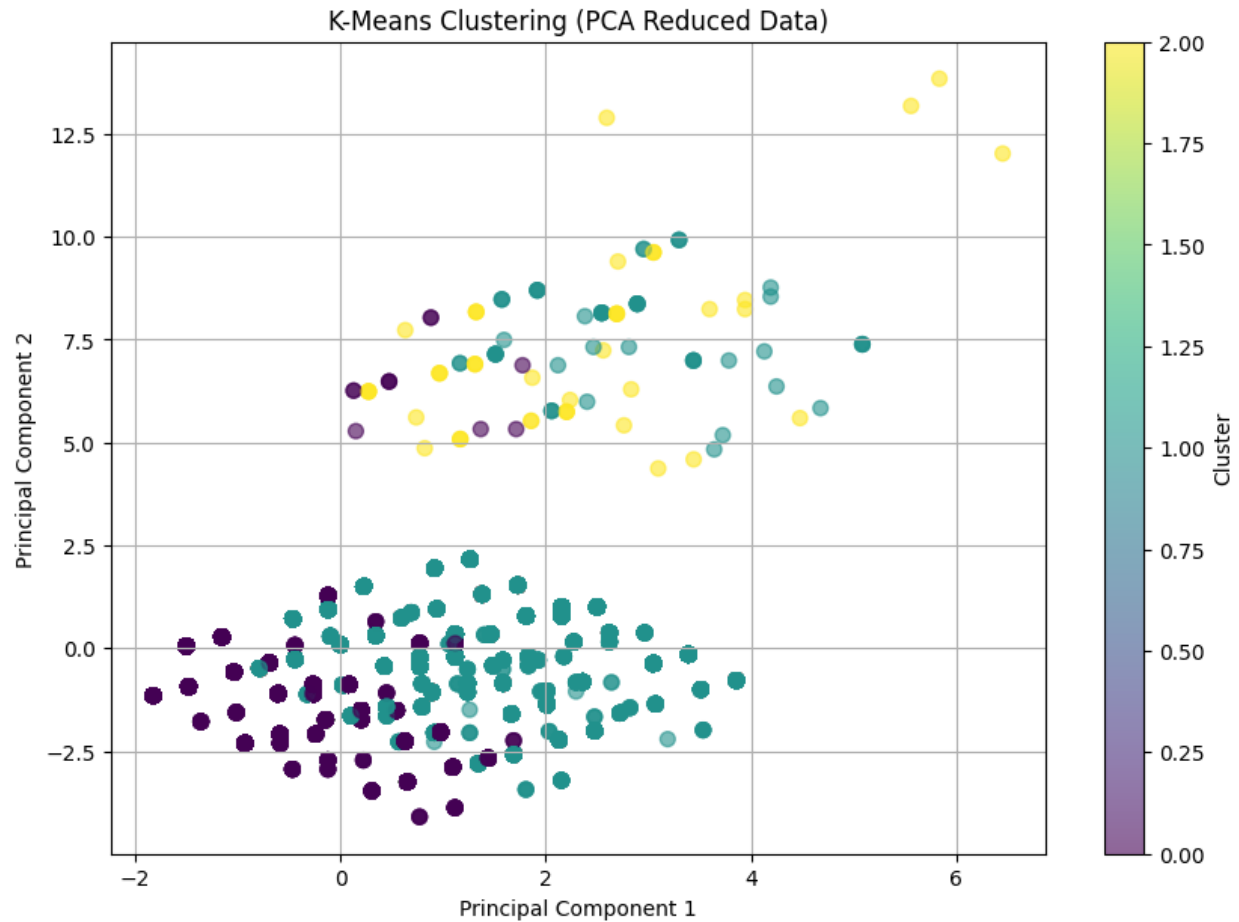


Figure: K-means clustering

From the visual structure of the clusters, there is clear cluster separation. The top cluster (yellow, Cluster 2) is well separated vertically from the lower two. This suggests that K-Means has successfully identified a distinct group in the data, especially along Principal Component 2. Overall, the clusters are well-formed. There is no major noise or dispersion; most points are near cluster centers. Cluster shapes are relatively compact, which aligns with K-Means' assumptions.

Cluster	Count	Total	Prevalence (%)
0	1047	3648	28.70
1	736	6346	11.60
2	2	6	33.33

Table: Diabetes Prevalence by Cluster (K-means Clustering Results)

DBSCAN Clustering

A K-distance graph was used to determine an appropriate value for the eps parameter in DBSCAN. The graph was generated using the distance to each point's 25th nearest neighbor ($\text{min_samples} = 25$), and a clear “elbow” was observed around the distance value of approximately 1.7, indicating a natural cutoff for cluster density. Based on this, DBSCAN clustering was performed using an epsilon (eps) value of 1.7 and a minimum sample threshold of 25, which effectively defined the density-based structure of the clusters while limiting the number of noise points.

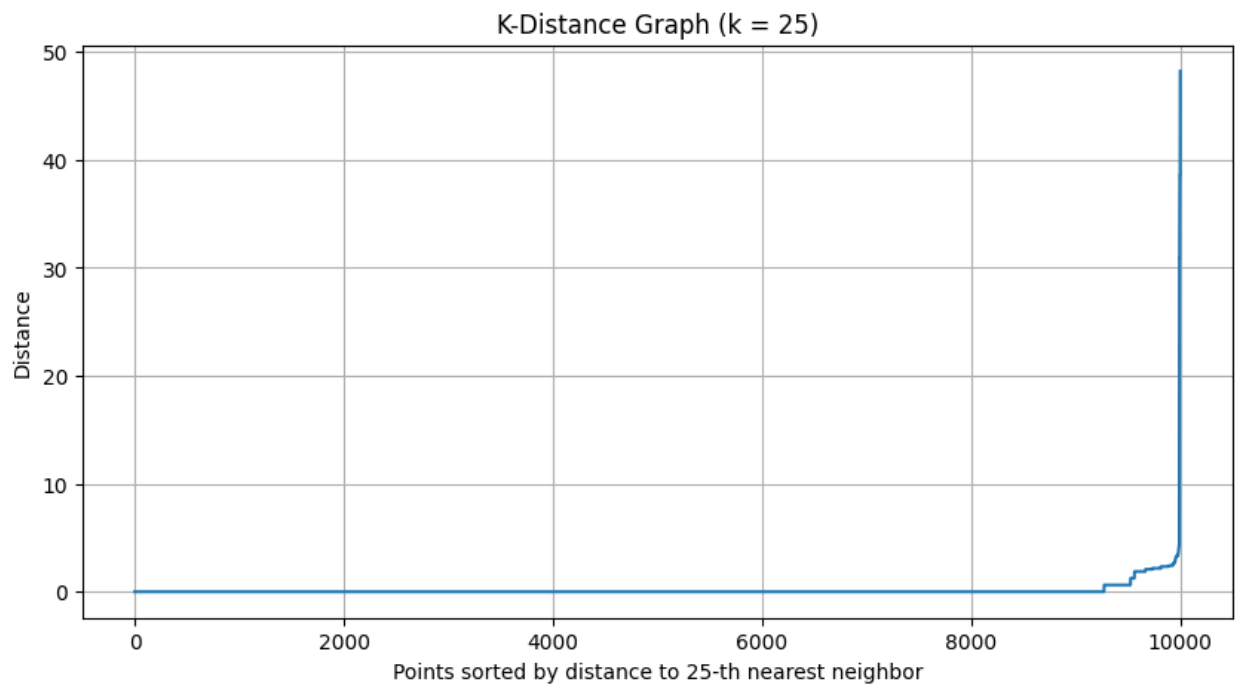


Figure : K-distance graph

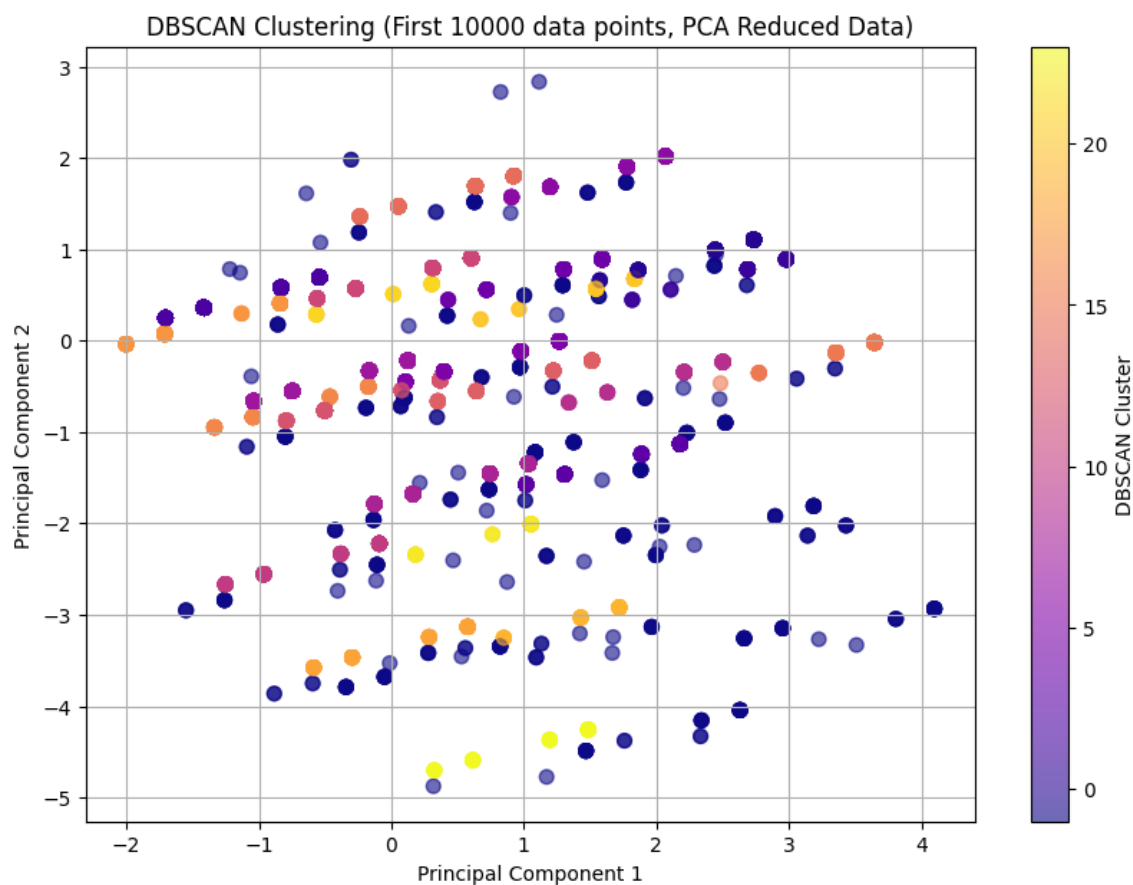


Figure: DBSCAN clusters in 2D using PCA

Cluster	Count	Total	Prevalence (%)
0	-1	372	68
1	0	257	94
2	1	100	43
3	2	3336	385
4	3	280	60
5	4	248	88
6	5	461	68
7	6	372	128
8	7	1398	206
9	8	569	88
10	9	111	47
11	10	177	19
12	11	577	113
13	12	341	69

14	13	125	33
15	14	482	134
16	15	190	77
17	16	106	6
18	17	199	9
19	18	78	4
20	19	38	4
21	20	64	31
22	21	38	3
23	22	35	2
24	23	46	6

Table: Diabetes Prevalence by Cluster (DBSCAN Clustering Results)

This prevalence analysis reveals clear variability in diabetes risk across DBSCAN clusters. A few small clusters (e.g., 1, 9, 15, 20) show very high prevalence rates, possibly representing high-risk behavioral or demographic groups. In contrast, several larger clusters (e.g., 2, 7, 8) have low to moderate prevalence, indicating more general populations with lower diabetes risk.

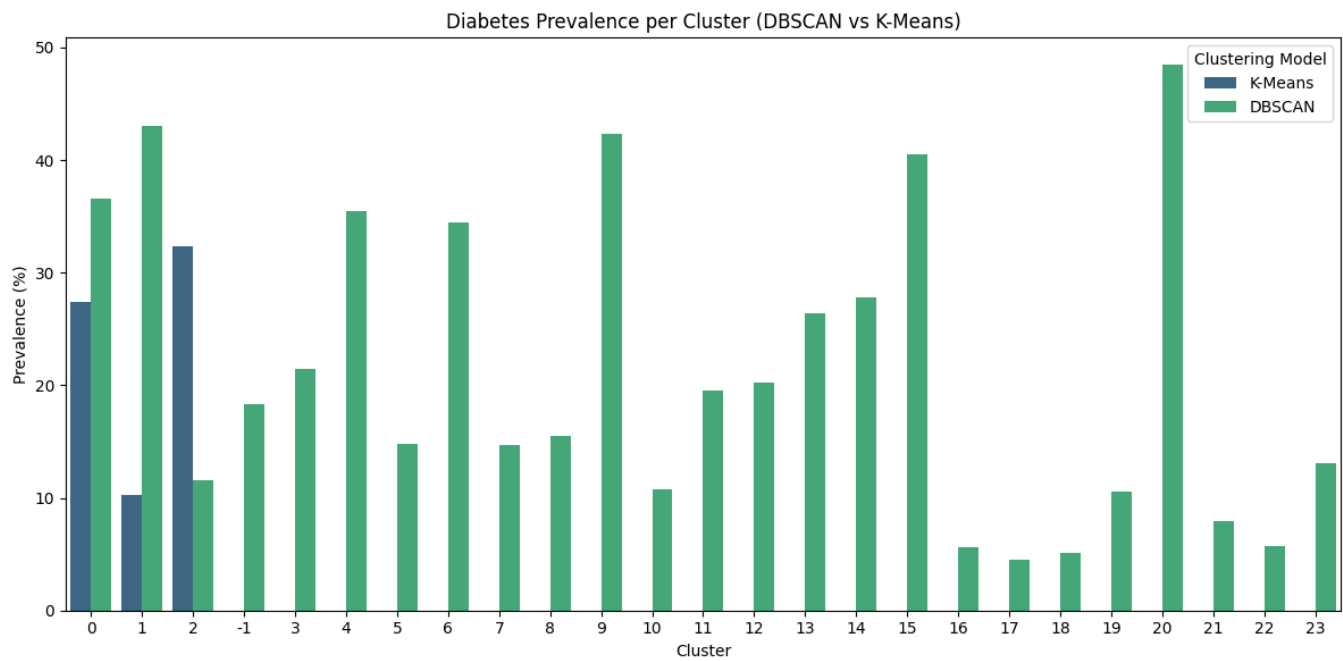


Figure: Comparison of Diabetes Prevalence Across Clusters (Insight 1)

The bar chart compares diabetes prevalence (%) across clusters (0–23) generated by DBSCAN and K-Means. K-Means clusters show a more uniform distribution of prevalence, suggesting balanced groupings, while DBSCAN reveals sharper contrasts, some clusters have very high or

low prevalence, highlighting potential high-risk subgroups or outliers. This implies DBSCAN may better identify extreme-risk populations for targeted interventions, whereas K-Means provides a broader overview.

4.2 Insight 2: Prevalence of Diabetes in the Population

To analyze the overall prevalence of diabetes in the population, we selected 10 features that broadly represent individual health status, lifestyle behaviors and demographic factors. These attributes are essential for identifying patterns in how diabetes is distributed across different population segments. The selected columns used in this analysis are HighBP, HighChol, BMI, Smoker, PhysActivity, Fruits, Veggies, GenHlth, Age and Income. These attributes provide a comprehensive snapshot of individual risk profiles and allow for meaningful segmentation.

K-Means Clustering

Using the Elbow Method, the optimal number of clusters was determined and the data was partitioned accordingly. The resulting clusters showed clear separation in the PCA-reduced space:

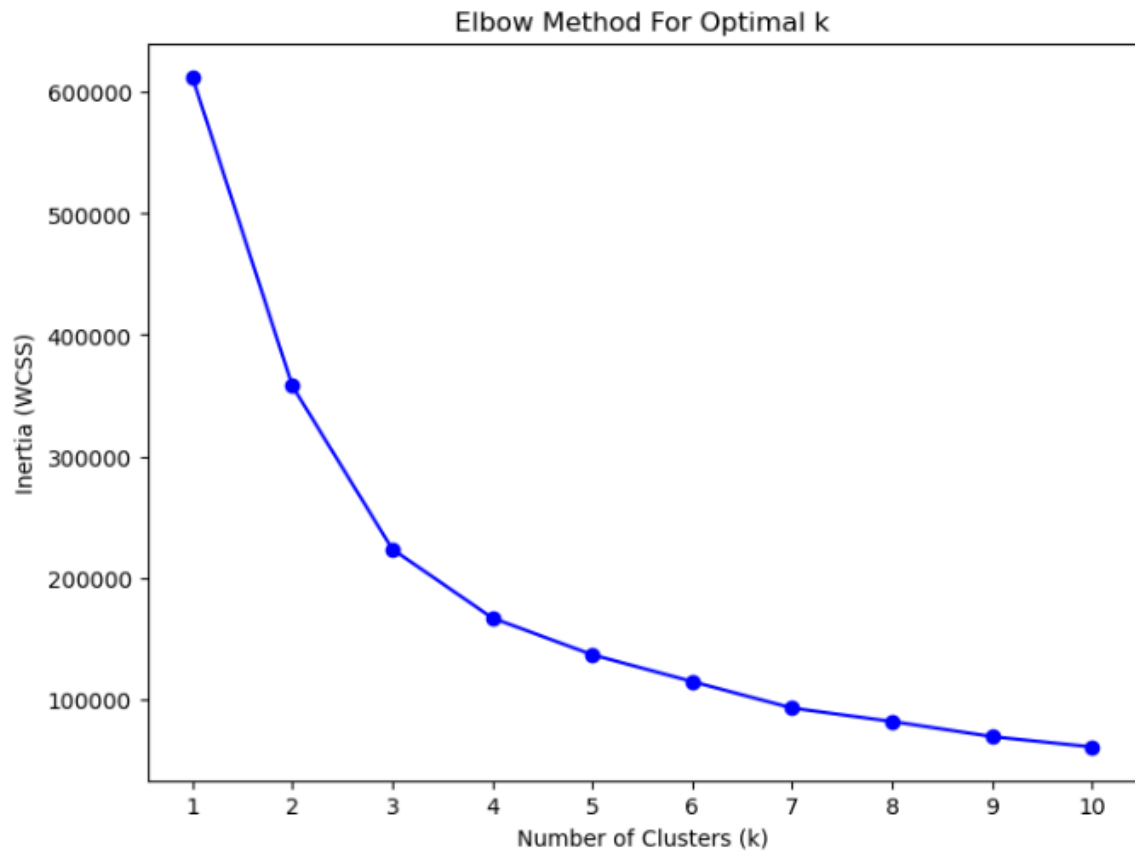


Figure: Elbow Method for Optimal k

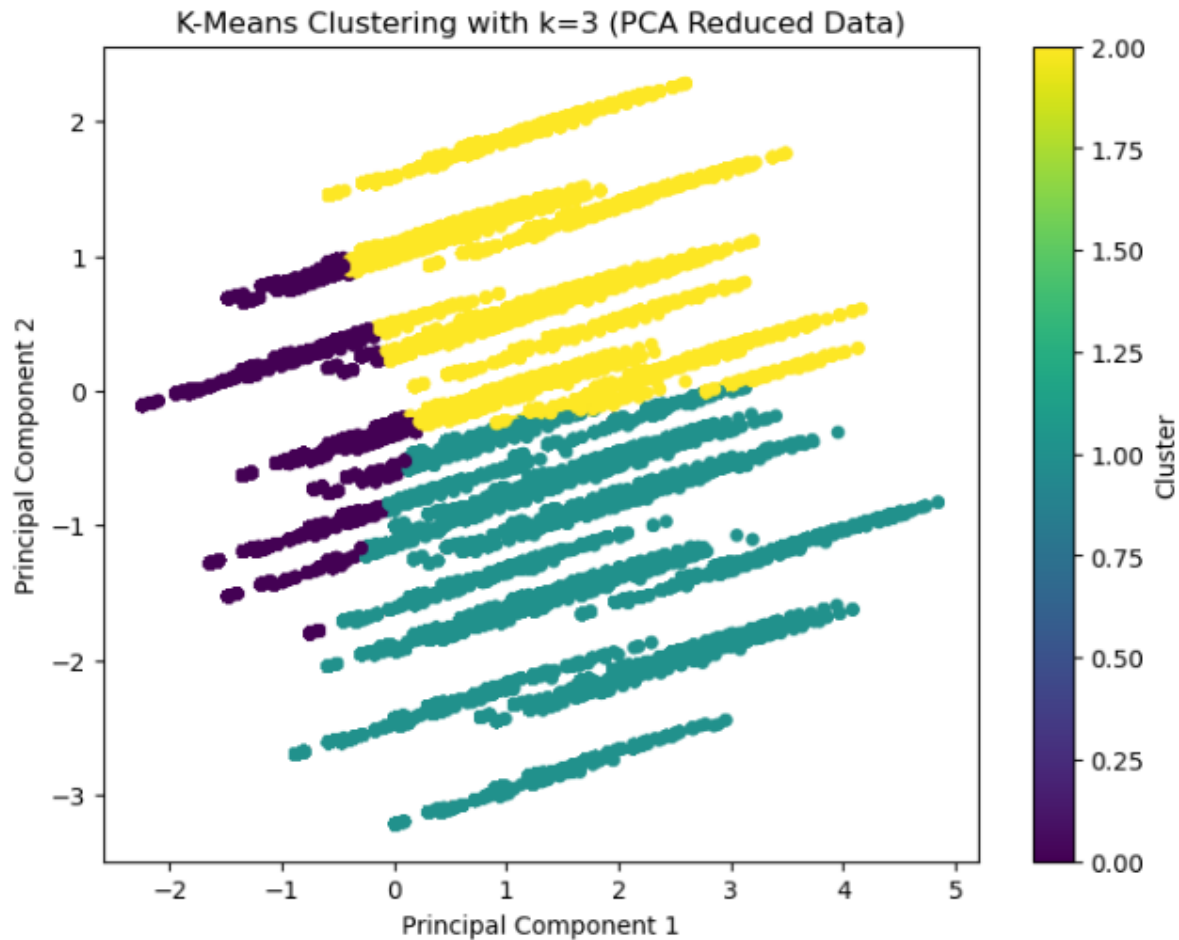


Figure: K-Means Clustering with k=3 (PCA Reduced Data)

Cluster	Count	Sum	Prevalence (%)
0	4738	90365	5.24
1	7921	41868	18.92
2	19234	70016	27.47

Table: K-Means Clustering Result

The K-Means results reveal that Cluster 0, with the largest total count, has the lowest diabetes prevalence ($\sim 5\%$), suggesting this group generally maintains healthier profiles. In contrast, Cluster 2 demonstrates the highest prevalence ($\sim 27.5\%$), indicating a substantial segment of the population with elevated risk factors. This broad segmentation helps policymakers and healthcare providers understand how diabetes is distributed across general population segments and highlights the importance of targeting resources toward higher-risk groups.

DBSCAN Clustering

DBSCAN was applied to the same features to identify natural clusters based on density, capturing more complex and nuanced groupings:

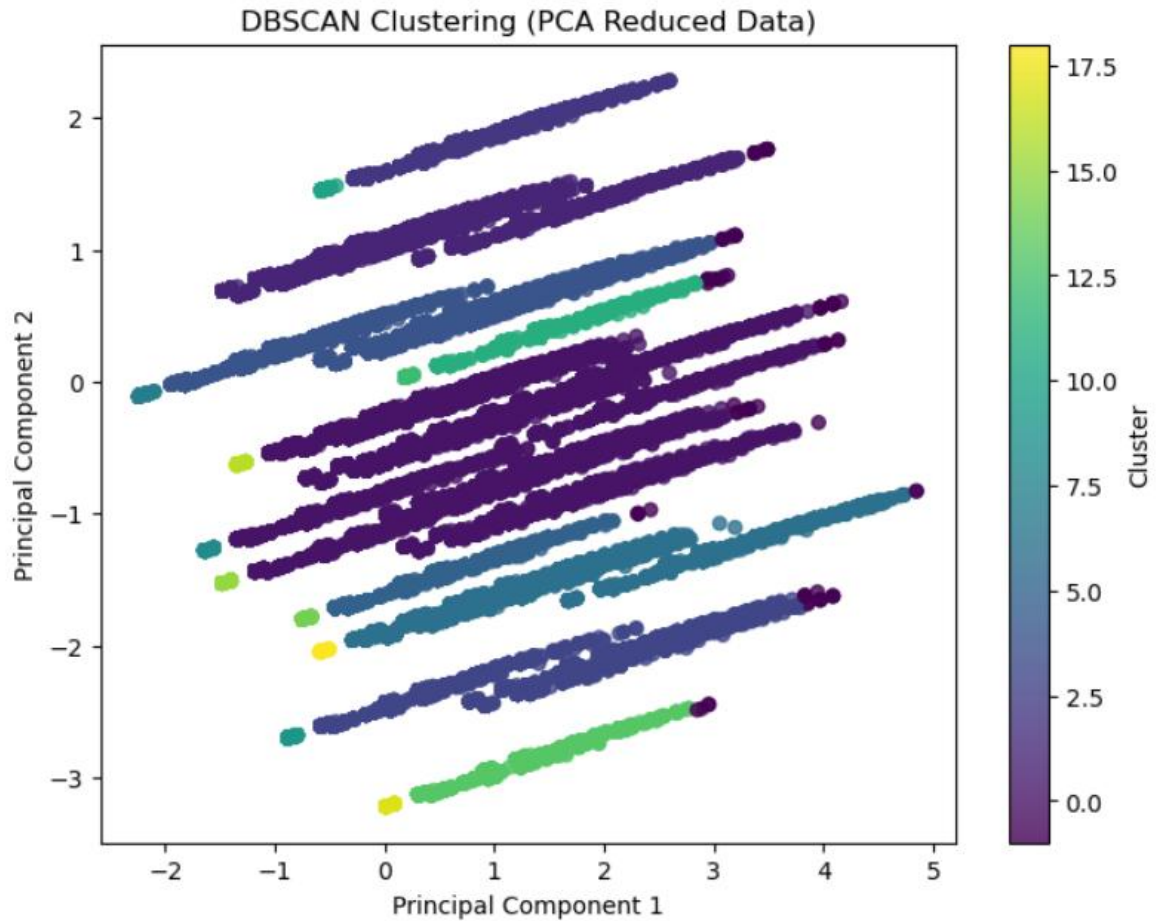


Figure: DBSCAN Clustering (PCA Reduced Data)

Cluster	Count	Sum	Prevalence (%)
-1	38	82	46.34
0	9139	54166	16.87
1	6606	38183	17.30
2	5464	18863	28.97
3	1011	7934	12.74
4	5648	48296	11.69
5	355	4096	8.67
6	2303	9357	24.61
7	86	9805	0.88
8	28	2525	1.11
9	8	750	1.07

10	79	956	8.26
11	851	2545	33.44
12	12	68	17.65
13	231	2621	8.81
14	6	377	1.59
15	8	637	1.26
16	14	658	2.13
17	0	220	0.00
18	6	110	5.45

Table: DBSCAN Clustering Result

Unlike K-Means, DBSCAN produced many clusters of varying sizes and detected small high-prevalence pockets as well as outliers (Noise, Cluster – 1) with an exceptionally high diabetes prevalence (46.34%). This demonstrates DBSCAN’s strength in revealing both dense risk area and isolated high-risk individuals who may not fit into larger population groups.

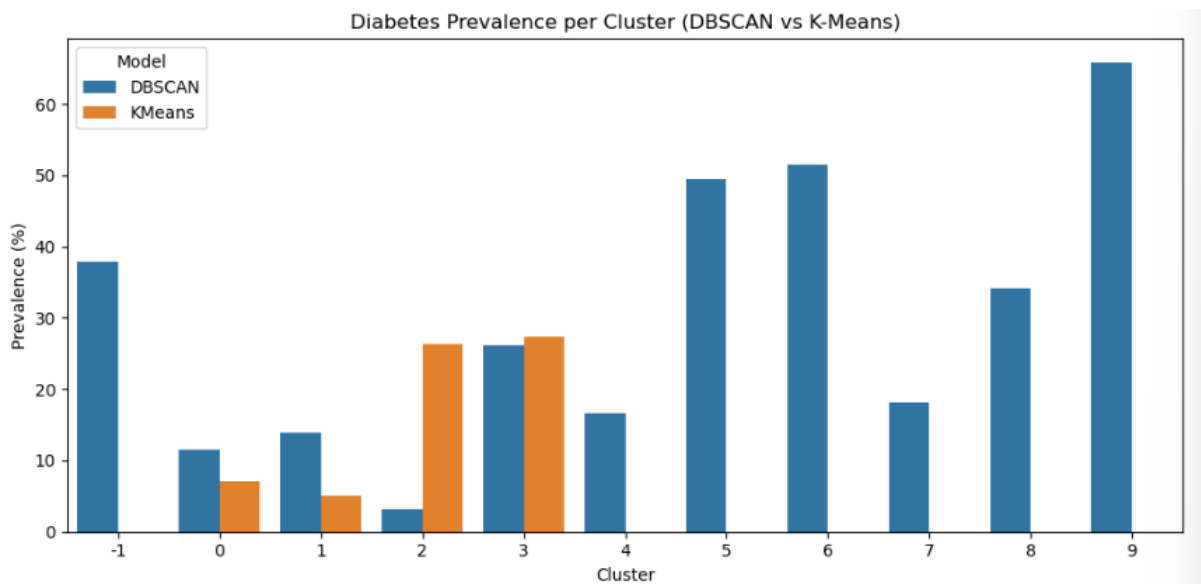


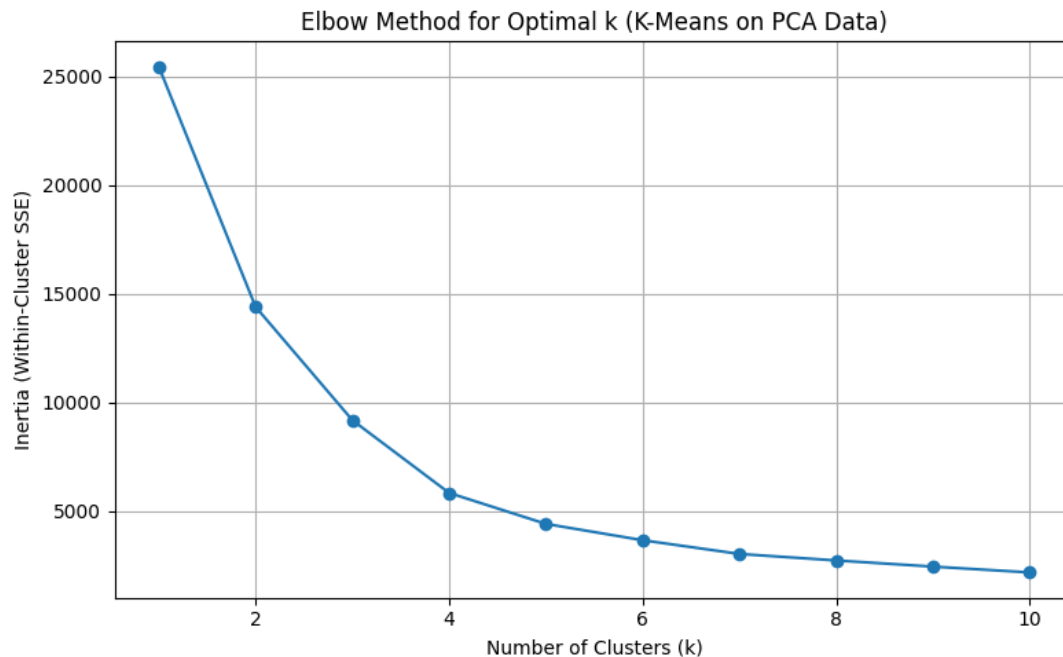
Figure: Comparison of Diabetes Prevalence Across Clusters (Insight 2)

Combining both clustering perspectives provides valuable understanding of how diabetes is distributed which K-Means offers broader, well-seperated population segments, ideal for general awareness and large scale prevention strategies while DBSCAN uncovers hidden, smaller high-risk pockets and outlier groups, enabling more targeted interventions for individuals who might be overlooked by general segmentation. This dual approach reinforces the importance of tailored public health measures. By recognizing that some segments have disproportionately higher

diabetes prevalence, health authorities can prioritize targeted screening, awareness campaigns and resource allocation to reduce diabetes incidence and improve population health outcomes.

4.3 Insight 3: Predictive Modeling for At-Risk Individuals

K-Means Clustering



The PCA scatter plot for K-Means shows how the data was grouped into four clusters. Each point represents an individual, plotted in the 2D PCA space. The clusters are well-separated in PCA space, which suggests that the selected features are effective at segmenting individuals by health risk. K-Means perform well for evenly distributed groups, making it suitable for general segmentation tasks. However, it does not isolate extreme outliers, potentially missing smaller high-risk pockets.

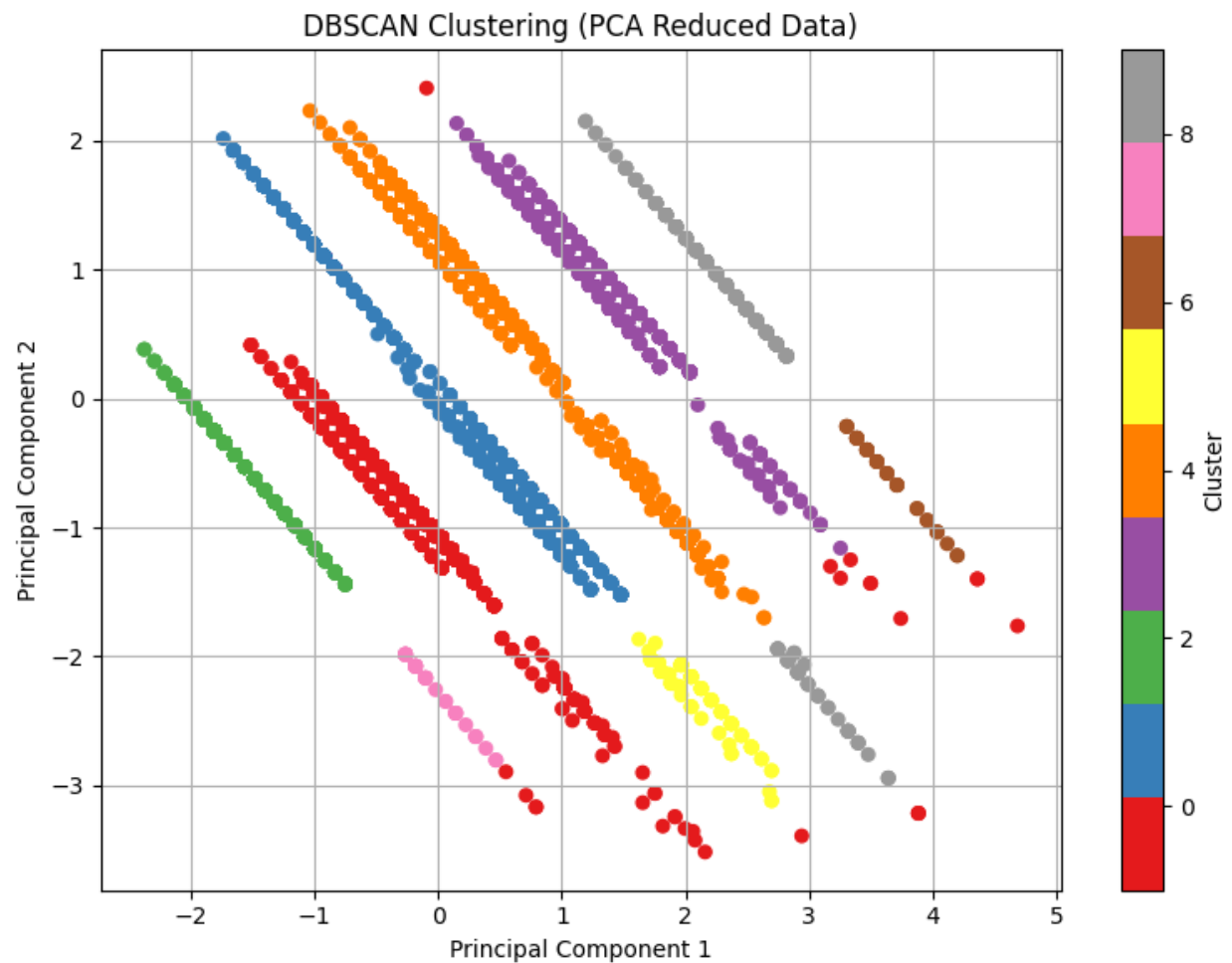


The table below summarizes the results of the K-Means clustering algorithm applied to PCA-reduced health risk features. The dataset was clustered into four groups ($k=4$), with the diabetes prevalence calculated within each cluster. Cluster 2 and 3 show significantly higher diabetes prevalence (above 25%), indicating that individuals grouped into these clusters are at higher risk. Cluster 0 and 1 represent lower risk groups with a prevalence under 10%. This segmentation helps in general population risk profiling, which is useful for broad preventive strategies.

Cluster	Count	Total	Prevalence (%)
0	164	2336	7.02
1	171	3394	5.04
2	604	2297	26.30
3	539	1973	27.32

Table: K-Means

DBSCAN Clustering

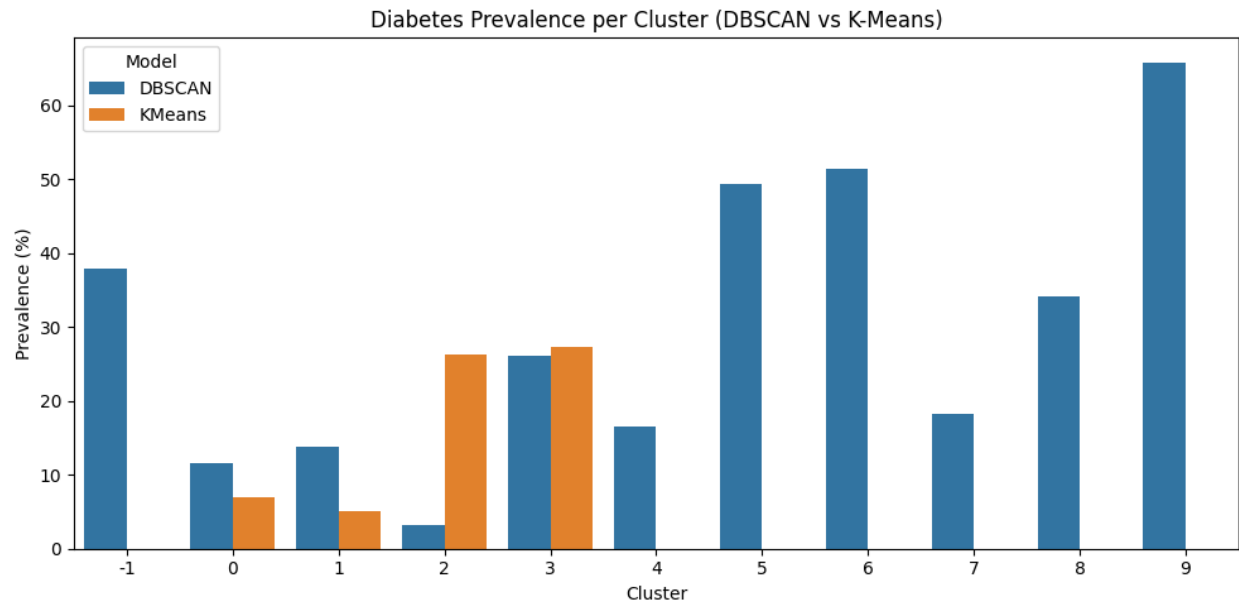


The DBSCAN clustering algorithm identified several clusters of varying density, including noise points (Cluster -1). The diabetes prevalence per cluster is summarized below:

Cluster	Count	Total	Prevalence (%)
-1	11	29	37.93
0	213	1843	11.56
1	322	2321	13.87
2	69	2154	3.22
3	345	1321	26.12
4	291	1755	16.58
5	42	85	49.41
6	18	35	51.43
7	6	33	18.18
8	134	392	34.18
9	27	41	65.85

Table: DBSCAN

DBSCAN successfully discovered small but highly concentrated high-risk clusters such as clusters 5, 6 and 9, all with diabetes prevalence exceeding 45%. The noise group (Cluster -1) also showed elevated prevalence (37.93%), indicating that even individuals not assigned to a cluster may be at risk. These findings are particularly valuable for early identification of at-risk individuals who may be overlooked by more rigid clustering algorithms.



The bar chart compares the diabetes prevalence rate in each cluster produced by K-Means and DBSCAN. While K-Means produced four broader clusters with moderate variation in prevalence (ranging from 5% to 27%), DBSCAN revealed a more granular and diverse grouping, some with extremely high prevalence rates exceeding 50% like clusters 5,6,9. The DBSCAN noise group (cluster -1) exhibited a notably high prevalence (37.93%), highlighting that individuals not grouped into any core cluster still represent a significant health risk. Overall, this comparison shows that DBSCAN offers better sensitivity for detecting high-risk individuals, whereas K-Means provides a general population segmentation.

5.0 Interpretation and discussion of results

The analyses conducted using K-Means and DBSCAN clustering provide meaningful insights into the prevalence of diabetes in the population, the impact of lifestyle factors on health outcomes and the identification of at risk individuals. This section synthesizes the results, compare the algorithms and discusses the implications for public health interventions and future decision making.

Insight 1: Impact of Lifestyle factors on Health Outcomes

The clustering of lifestyle-related features (such as physical activity, diet and alcohol consumption) revealed clear patterns which **K-Means** identified three broad clusters with relatively distinct lifestyle profiles. The diabetes prevalence ranged from ~11% to ~33%, showing how different combinations of lifestyle behaviors correlate with health outcomes while **DBSCAN** produced a larger number of clusters, capturing subtle variations in behavior and risk. It highlights that some smaller groups have a significantly higher diabetes prevalence (>35%), which K-Means did not isolate.

Key takeaway: Lifestyle factors play a critical role in chronic disease risk. DBSCAN's finer granularity suggests that small pockets of the population engage in high-risk behaviors that may not be evident in broader segmentation. Public health programs should therefore not only address general trends but also target smaller high-risk communities with specific interventions like smoking cessation, fitness incentives and nutrition education.

Insight 2: Prevalence of Diabetes in the Population

Using the selected 10 features covering health status, behaviors and demographics clustering revealed important population segments which **K-Means** formed three main clusters with diabetes prevalence ranging from ~5% to ~27%. This highlights that while most of the population falls into low-to-moderate risk categories, a sizeable segment shows considerably higher prevalence while **DBSCAN** produced many smaller cluster and identified isolated groups and outliers with very high prevalence, including a noise group with a prevalence above 40%.

Key takeaway: While K-Means provides an overall picture suitable for general resource allocation, DBSCAN demonstrates that hidden high-risk subgroups exist within the population. This suggests that a one-size-fits-all approach is insufficient; policymakers should combine broad strategies with more granular outreach to ensure these vulnerable groups are not overlooked.

Insight 3: Predictive Modeling or At-Risk Individuals

The clustering of key predictive health features illustrates the potential for early risk identification which **K-Means** created clear segments that help differentiate between low and high-risk individuals. Clusters with higher prevalence (~26% - 27%) can be prioritized for preventive care while **DBSCAN** revealed smaller, dense groups with extremely high prevalence rates (>45%), plus noise clusters that might otherwise be missed. These findings emphasize the importance of models that detect outliers.

Key takeaway: Predictive segmentation enables proactive identification of individuals at elevated risk of diabetes. Combining K-Means and DBSCAN ensures that both large at-risk groups and rare, high-risk cases are flagged for early intervention, which can greatly enhance the efficiency of healthcare screening and personalized recommendations.

Overall Discussion

The comparative results highlight the complementary strengths of the two clustering approaches:

- **K-Means** is effective for general population profiling when clusters are expected to be spherical and evenly distributed. Its simple structure and interpretability make it practical for large-scale policy planning and general outreach.
- **DBSCAN**, with its density-based clustering, excels at discovering non-linear, irregular patterns and isolating noise points. This makes it invaluable for uncovering hidden high-risk segments that traditional methods may miss.

By combining these methods, stakeholders gain a more comprehensive understanding of diabetes distribution, risk factors and behavioral patterns across the population. This dual approach strengthens evidence-based decision-making and allows public health authorities to design targeted, tiered interventions that maximize the impact of prevention and education programs.

6.0 Conclusion

This project demonstrates the practical value of applying unsupervised clustering techniques to large-scale health data for uncovering meaningful patterns in the prevalence of diabetes and related risk factors. By leveraging both K-Means and DBSCAN algorithms, we successfully segmented the population into distinct groups based on health status, lifestyle behaviors and demographic attributes.

The combination of distance-based and density based clustering approaches highlighted the importance of using multiple methods to capture both general trends and hidden patterns within the data. K-Means provided clear, well separated clusters suitable for broad population profiling, while DBSCAN revealed smaller, more nuanced pockets of high-risk individuals and outliers that might otherwise go unnoticed.

Overall, these findings reinforce how data mining and clustering can support evidence-based public health decision-making. By identifying population segments with higher diabetes prevalence and understanding how health behaviors cluster within communities, stakeholders can design more targeted interventions, optimize resource allocation and improve early detection and prevention strategies.

Beyond technical implementation, this work highlights the broader role of data-driven analysis in addressing real-world health challenges. It demonstrates how combining robust data preprocessing, thoughtful feature selection and appropriate algorithm selection can turn raw data into actionable knowledge. These insights lay a strong foundation for future research and practical applications in predictive health analytics, personalized wellness programs and continuous health monitoring.

In conclusion, clustering techniques offer an accessible yet powerful way to extract valuable insights from complex health data, ultimately contributing to better health outcomes and more informed, effective public health strategies.

Project Reflection – WOO CHENG SHUAN

(A23CS0283)

For this project, the BRFSS 2015 dataset was analyzed using both classification and clustering algorithms to glean valuable insights. In Project 1, we investigated topics including diabetes prevalence, lifestyle influence and gender-based risk through supervised learning with algorithms like Random Forest, XGBoost and KNN. Data cleansing, feature selection, model training and assessment using performance metrics and visualizations were among the tasks. In Project 2, we shifted to unsupervised learning using K-Means and DBSCAN to find hidden patterns and clusters within the population, particularly in relation to diabetes risk and health habits. Prevalence among cluster was computed for interpretation and PCA was utilized for visualization.

One of the main challenges was generating evaluation figures using Python, especially as I didn't know much this course about coding. After feeling overpowered at first, I eventually learned how to created charts such as ROC curves and confusion matrices which improved my comprehension of model performance from the Internet. This hands-on experience deepened my understanding of machine learning principles and I feel more comfortable using Python for data analysis. Overall, the project enhanced my appreciations for the data mining process and provided valuable analytical skills for future tasks.

Course Reflection – WOO CHENG SHUAN

(A23CS0283)

I obtained a solid foundation in data mining from theory to real-world application thanks to the eye-opening course SECP2753. I gained knowledge about handling data pre-processing, applying classification and clustering techniques and using association rule mining to find patterns through lectures, assignments and projects. It was simpler to understand how these ideas function in practical situations by using programs like Excel and Python.

By examining data mining procedures, developing solutions using supervised and unsupervised learning and putting forward data-driven business potential through case studies and reports, I think I achieved the course learning objectives. Along the way, I strengthened soft skills like teamwork, report writing and results presentation while also honing my technical abilities and gaining experience working with actual data.

I'm motivated to continue learning because of this course. I intend to investigate Python and packages to enhance these abilities. All things considered, SECP2753 has given me a strong foundation for a career in data science by increasing my confidence and preparing me for employment in data analysis and AI in the future.